Equity and Justice in Critical Mineral Supply Chains

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Key Questions

1. How can we ensure the energy transition addresses global climate justice and the Global South doesn't bear the brunt of the transition?
2. What are global forces impacting environmental justice and supply chains?
3. What could be GPIs role in addressing questions of social equity in global supply chains?

Why are we talking about this?

“Imagine a mountain of gravel and stone just avalanching down on people, crushing legs and arms, spines. I met people whose legs had been amputated, who had metal bars in where their legs used to be.”



Environmental Justice Concerns at Home

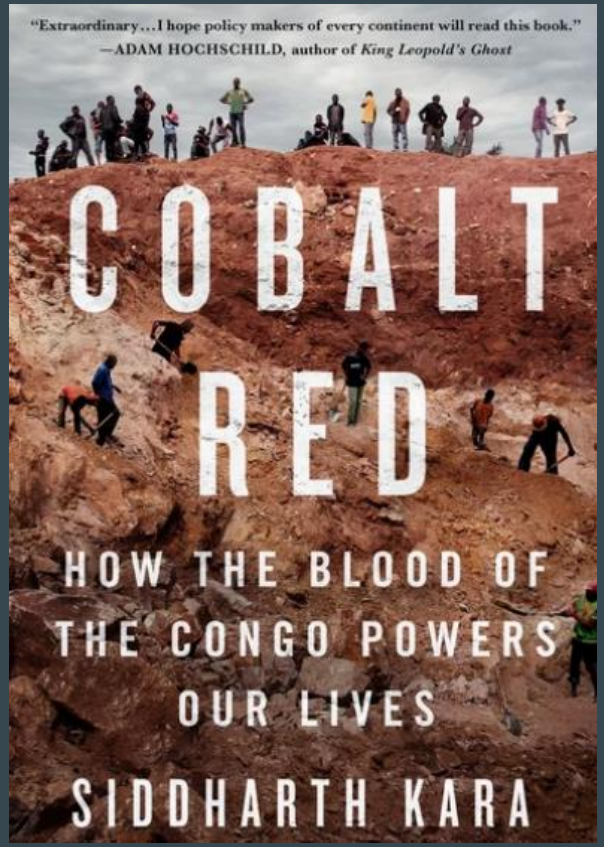


"In the energy towns of Arkansas, the coming lithium rush is bringing with it the risk of repeating the same mistakes and inequities of the past."

Source: Grist

Key issues

- Human rights abuses
  - Forced labor (e.g., Uyghurs)
  - Child labor
  - Lack of labor protections
  - Land displacement
- Environmental damages
  - Critical mineral mining for solar and EVs
- The meaning of a “Just” Transition



How Critical Minerals Connect to GPI's Work

- Solar Panels: high volumes of aluminum and copper and low volumes of cobalt, nickel, lithium, rare earth elements, chromium, zinc
- Electric Vehicles: lithium, cobalt, manganese, nickel, and graphite
- CO_2 utilization to make mining less energy intensive
- Considerations: water stress, pollution from mining, increased energy for extraction/processing, worker safety, labor exploitation



Source: National Wildlife Federation

Source: UT Austin

Sources of Critical Minerals and Rare Earth Elements



Geopolitics of Critical Minerals and Rare Earth Elements

- The United States is 100% import dependent on 12 critical minerals
  - More than 50% dependent on 31 others
- China is the largest source of of these imports for 26/51 critical minerals
- U.S strategy: onshore, nearshore, and friend-shore supply chains
  - China views this as a containment strategy
- Consequences of escalation:
  - Supply chain disruptions of REE and critical minerals hurt defense industry & energy transition.
  - Prohibiting export of technologies for REM refining
  - Threats to environmental and labor regulations, particularly in the Global South.

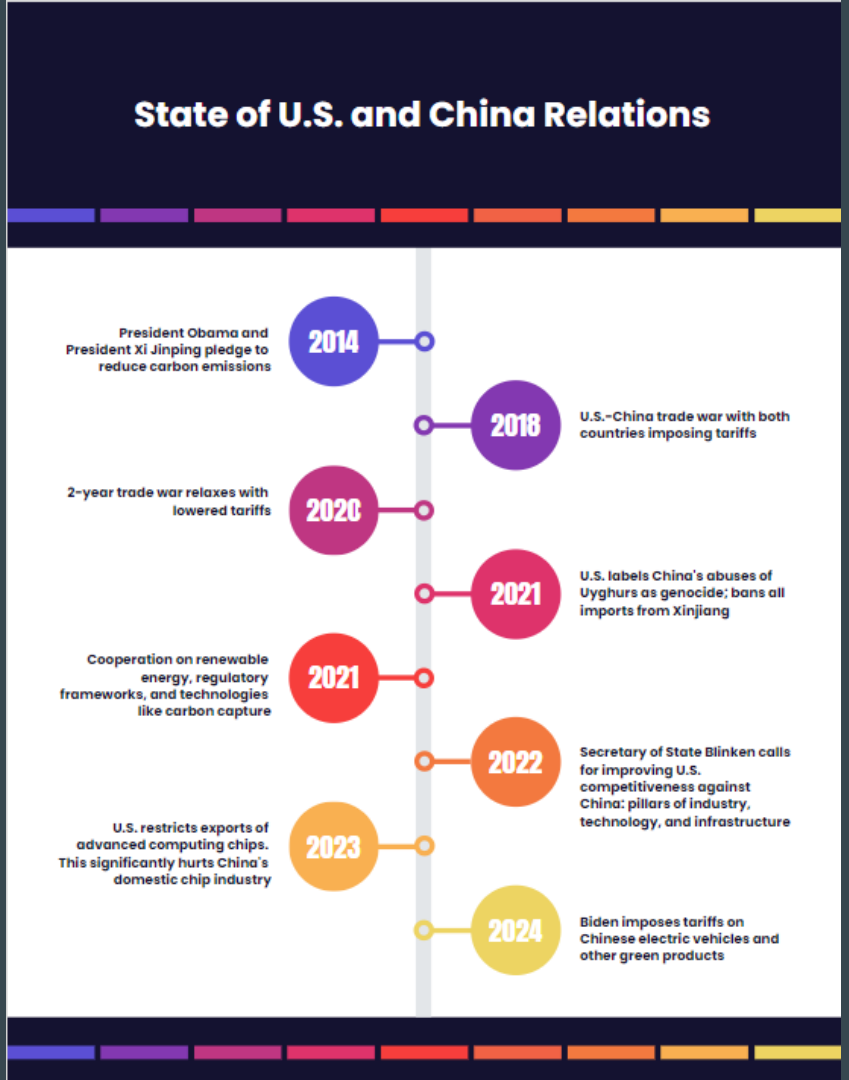
Source: Council on Foreign Relations

U.S. Bilateral Agreements

- U.S.-Australia Critical Minerals Partnership - signed in 2019
- U.S.-Canada Joint Action Plan on Critical Minerals - launched in 2020
- U.S.-EU Trade and Technology Council - launched in 2021
- U.S.-Japan Critical Minerals Cooperation - signed in 2023

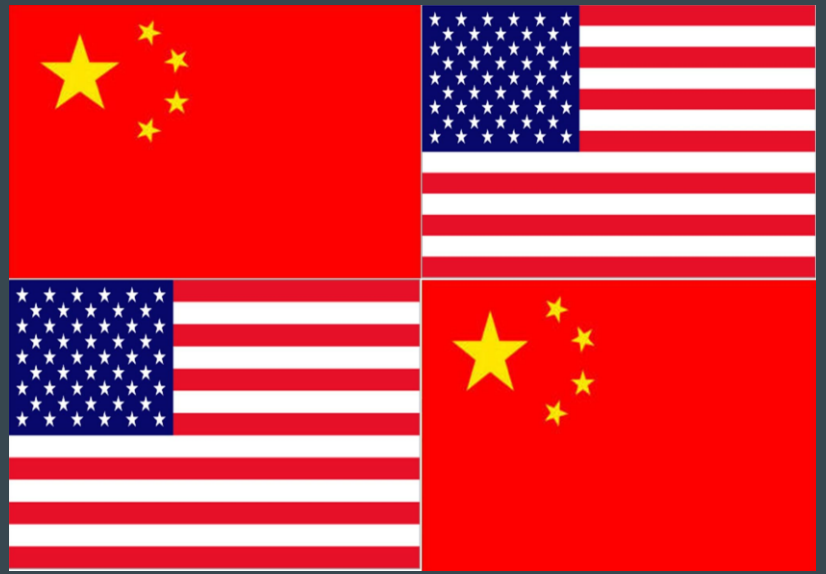
Sources: White House, IEA, U.S. Department of State, U.S. Trade Representative

Key Moments in U.S and China Relations



Big Takeaways: U.S and China Relations

- China's dominance of clean energy supply chains is massive
  - Strategic advantage in processing capacity
- U.S is caught between climate goals and competitive advantage
- Access to transparent data around supply chains is minimal
- Rise of industrial policy and protectionist positions weakens the role of international institutions



Thank you to Rai Cai from the Center for Strategic and International Studies

What is the path forward?

- Build domestic manufacturing capacity
- Strengthen international relationships to avoid supply shocks
- Prioritize labor rights across the supply chain
- Develop a competitive edge by investing in technology and innovation

What Role Can GPI Play?

Research

GPIs proposed Equity Hub could compile research from human rights experts and conduct our own to ensure practices across teams consider EJ impacts on the Global South.

Questions to investigate:

- What are the implications of focusing on a domestic supply chain to the clean energy transition?
- Who benefits from an emphasis on human well-being in supply chains? Who suffers?

Education

GPI can use research findings to inform (internal) community of these issues through “lunch n’ learns,” all-staff presentations, newsletters, and other venues.

Coalition Building

GPI can support, encourage, and educate partner organizations on the global social justice issues associated with the clean energy transition and alternatives for sourcing from equitable supply chains

State and Federal Advocacy

GPI can work with government agencies to support the development of programs and policies that incentivize American manufacturing and materials sourcing with strong labor/environmental standards.

Recap

- Critical minerals and rare earth elements are essential
- Lots of abuse, exploitation, and harm in these supply chains
- U.S found a lot of lithium in Arkansas. Like a lot.
- China dominates the critical mineral/rare earth element processing industry
- Group activity: what is the path forward for equitable supply chains?

Breakout Rooms

Resourcing the Energy Transition: Principles to Guide Critical Energy Transition Minerals Towards Equity and Justice

Instructions:

- One person in your breakout room should make a copy of this template
- Create a shareable link that anyone can edit (go to share at the top right, click under general access, change the setting to Anyone with the link, and on the right hand side change from “viewer” to “editor”)
- Share the link with your breakout room participants so everyone can work on the same document
- After completing the table, pick a facilitator and discuss your findings!

Resources

- https://www.nwf.org/-/media/Documents/PDFs/NWF-Reports/2022/Critical-Minerals-for-Clean-Energy-Reference-Guide.pdf
- https://apnews.com/article/china-rights-minerals-renewable-abuse-fa0699bddd5ac076377ddedbd20fac50
- https://www.state.gov/wp-content/uploads/2022/07/Forced-Labor-and-the-Clean-Energy-Transition-Finding-A-Responsible-Way-Forward.pdf
- https://grist.org/energy/deja-vu-comes-to-arkansas-as-lithium-follows-oil/?utm_medium=email
- https://www.cfr.org/timeline/us-china-relations
- http://cfr.org/blog/how-secure-critical-minerals-clean-energy-without-alienating-china